is given by its projection onto the z-axis, which can be seen from the figure to be $F\cos\phi$. The distance from the mass $\delta\Delta V$ to the unit mass at the origin is the spherical coordinate ρ. Therefore, the z-component of the force due to the small piece ΔV is

$$\begin{matrix}z\text{-component}\\ \text{of force}\end{matrix} = \frac{G(\delta\Delta V)(1)}{\rho^2}\cos\phi.$$

Adding the contributions of the small pieces, we get a vertical force with magnitude

$$\begin{aligned}F &= \int_0^{2\pi}\int_0^{\pi/2}\int_0^a \left(\frac{G\delta}{\rho^2}\right)(\cos\phi)\rho^2\sin\phi\,d\rho\,d\phi\,d\theta = \int_0^{2\pi}\int_0^{\pi/2} G\delta(\cos\phi\sin\phi)\rho\Big|_{\rho=0}^{\rho=a} d\phi d\theta\\ &= \int_0^{2\pi}\int_0^{\pi/2} G\delta a\cos\phi\sin\phi\,d\phi\,d\theta = \int_0^{2\pi} G\delta a\left(-\frac{(\cos\phi)^2}{2}\right)\Big|_{\phi=0}^{\phi=\pi/2} d\theta\\ &= \int_0^{2\pi} G\delta a\left(\frac{1}{2}\right)d\theta = G\delta a\pi.\end{aligned}$$

The integral in this example is improper because the region of integration contains the origin, where the force is undefined. However, it can be shown that the result is nevertheless correct.

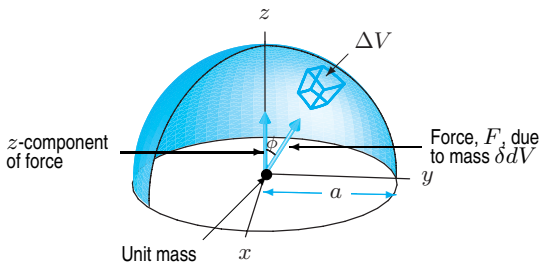


Figure 16.49: Gravitational force of hemisphere on mass at origin

Exercises and Problems for Section 16.5

Exercises

1. Match the equations in (a)–(f) with one of the surfaces in (I)–(VII).

(a) $x = 5$ **(b)** $x^2 + z^2 = 7$ **(c)** $\rho = 5$
(d) $z = 1$ **(e)** $r = 3$ **(f)** $\theta = 2\pi$

(I) Cylinder, centered on x-axis.
(II) Cylinder, centered on y-axis.
(III) Cylinder, centered on z-axis.
(IV) Plane, perpendicular to the x-axis.
(V) Plane, perpendicular to the y-axis.
(VI) Plane, perpendicular to the z-axis.
(VII) Sphere.

In Exercises 2–7, find an equation for the surface.

2. The vertical plane $y = x$ in cylindrical coordinates.

3. The top half of the sphere $x^2+y^2+z^2 = 1$ in cylindrical coordinates.

4. The cone $z = \sqrt{x^2 + y^2}$ in cylindrical coordinates.

5. The cone $z = \sqrt{x^2 + y^2}$ in spherical coordinates.

6. The plane $z = 10$ in spherical coordinates.

7. The plane $z = 4$ in spherical coordinates.

In Exercises 8–9, evaluate the triple integrals in cylindrical coordinates over the region W.

8. $f(x, y, z) = \sin(x^2 + y^2)$, W is the solid cylinder with height 4 and with base of radius 1 centered on the z axis at $z = -1$.

9. $f(x, y, z) = x^2 + y^2 + z^2$, W is the region $0 \le r \le 4$, $\pi/4 \le \theta \le 3\pi/4$, $-1 \le z \le 1$.

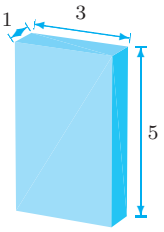
In Exercises 10–11, evaluate the triple integrals in spherical coordinates.

10. $f(\rho, \theta, \phi) = \sin\phi$, over the region $0 \le \theta \le 2\pi$, $0 \le \phi \le \pi/4$, $1 \le \rho \le 2$.

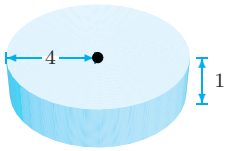
11. $f(x, y, z) = 1/(x^2 + y^2 + z^2)^{1/2}$ over the bottom half of the sphere of radius 5 centered at the origin.

For Exercises 12–18, choose coordinates and set up a triple integral, including limits of integration, for a density function f over the region.

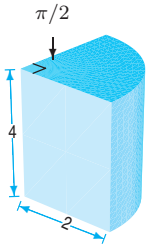
12.



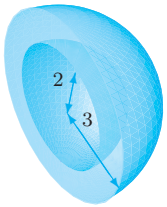
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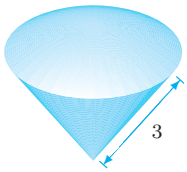
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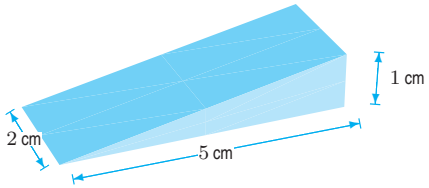
15.



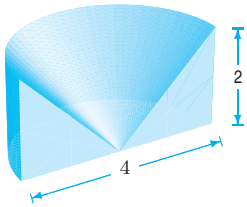
16. A piece of a sphere; angle at the center is $\pi/3$.



17.



18.



Problems

19. Write a triple integral in cylindrical coordinates giving the volume of a sphere of radius K centered at the origin. Use the order $dz\,dr\,d\theta$.

20. Write a triple integral in spherical coordinates giving the volume of a sphere of radius K centered at the origin. Use the order $d\theta\,d\rho\,d\phi$.

If W is the region in Figure 16.50, what are the limits of integration in Exercises 21–23?

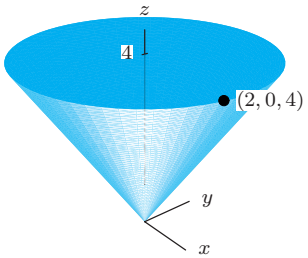


Figure 16.50: Cone with flat top, symmetric about z-axis

21. $\displaystyle\int_?^? \int_?^? \int_?^? f(r, \theta, z) r\,dz\,dr\,d\theta$

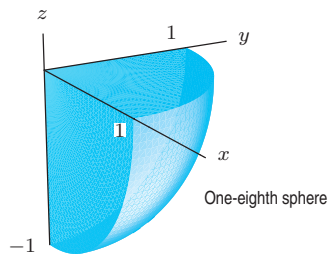
22. $\displaystyle\int_?^? \int_?^? \int_?^? g(\rho, \phi, \theta)\rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$

23. $\displaystyle\int_?^? \int_?^? \int_?^? h(x, y, z)\,dz\,dy\,dx$

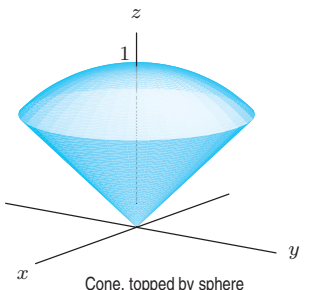
For the regions W shown in Problems 24–26, write the limits of integration for $\int_W dV$ in the following coordinates:

(a) Cartesian **(b)** Cylindrical **(c)** Spherical

24.

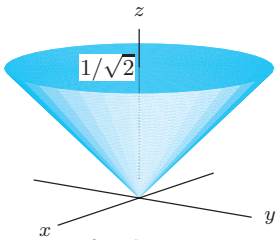


25.



Cone, topped by sphere of radius 1 centered at origin, 90° at vertex

26.



Cone, flat on top, $\pi/2$ at vertex

27. Write a triple integral representing the volume above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere of radius 2 centered at the origin. Include limits of integration but do not evaluate. Use:

(a) Cylindrical coordinates
(b) Spherical coordinates

28. Write a triple integral representing the volume of the region between spheres of radius 1 and 2, both centered at the origin. Include limits of integration but do not evaluate. Use:

(a) Spherical coordinates.
(b) Cylindrical coordinates. Write your answer as the difference of two integrals.

In Problems 29–34, write a triple integral including limits of integration that gives the specified volume.

29. Under $\rho = 3$ and above $\phi = \pi/3$.

30. Under $\rho = 3$ and above $z = r$.

31. The region between $z = 5$ and $z = 10$, with $2 \leq x^2 + y^2 \leq 3$ and $0 \leq \theta \leq \pi$.

32. Between the cone $z = \sqrt{x^2 + y^2}$ and the first quadrant of the xy-plane, with $x^2 + y^2 \leq 7$.

33. The cap of the solid sphere $x^2 + y^2 + z^2 \leq 10$ cut off by the plane $z = 1$.

34. Below the cone $z = r$, above the xy-plane, and inside the sphere $x^2 + y^2 + z^2 = 8$.

35. **(a)** Write an integral (including limits of integration) representing the volume of the region inside the cone $z = \sqrt{3(x^2 + y^2)}$ and below the plane $z = 1$.
(b) Evaluate the integral.

36. Find the volume between the cone $z = \sqrt{x^2 + y^2}$ and the plane $z = 10 + x$ above the disk $x^2 + y^2 \leq 1$.

37. Find the volume between the cone $x = \sqrt{y^2 + z^2}$ and the sphere $x^2 + y^2 + z^2 = 4$.

38. The sphere of radius 2 centered at the origin is sliced horizontally at $z = 1$. What is the volume of the cap above the plane $z = 1$?

39. Suppose W is the region outside the cylinder $x^2+y^2 = 1$ and inside the sphere $x^2 + y^2 + z^2 = 2$. Calculate

$$\int_W (x^2 + y^2)\, dV.$$

40. Write a triple integral representing the volume of a slice of the cylindrical cake of height 2 and radius 5 between the planes $\theta = \pi/6$ and $\theta = \pi/3$. Evaluate this integral.

41. Write a triple integral representing the volume of the cone in Figure 16.51 and evaluate it.

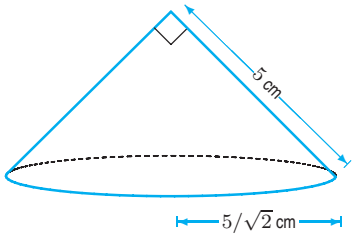


Figure 16.51

Without performing the integration, decide whether each of the integrals in Problems 42–43 is positive, negative, or zero. Give reasons for your decision.

42. W_1 is the unit ball, $x^2 + y^2 + z^2 \leq 1$.

(a) $\int_{W_1} \sin\phi\, dV$ **(b)** $\int_{W_1} \cos\phi\, dV$

43. W_2 is $0 \leq z \leq \sqrt{1 - x^2 - y^2}$, the top half of the unit ball.

(a) $\int_{W_2} (z^2 - z)\, dV$ **(b)** $\int_{W_2} (-xz)\, dV$

44. The insulation surrounding a pipe of length l is the region between two cylinders with the same axis. The inner cylinder has radius a, the outer radius of the pipe, and the insulation has thickness h. Write a triple integral, including limits of integration, giving the volume of the insulation. Evaluate the integral.

45. Assume p, q, r are positive constants. Find the volume contained between the coordinate planes and the plane

$$\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 1.$$

46. A cone stands with its flat base on a table. The cone's circular base has radius a; the vertex (tip) is at a height of h above the center of the base. Write a triple integral, including limits of integration, representing the volume of the cone. Evaluate the integral.

47. A half-melon is approximated by the region between two concentric spheres, one of radius a and the other of radius b, with $0 < a < b$. Write a triple integral, including limits of integration, giving the volume of the half-melon. Evaluate the integral.

48. A bead is made by drilling a cylindrical hole of radius 1 mm through a sphere of radius 5 mm. See Figure 16.52.

(a) Set up a triple integral in cylindrical coordinates representing the volume of the bead.
(b) Evaluate the integral.

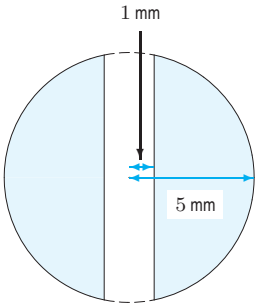


Figure 16.52

49. A pile of hay is approximately in the shape of $0 \le z \le 2 - x^2 - y^2$, where x, y, z are in meters. At height z, the density of the hay is $\delta = (2 - z)$ kg/m^3.

(a) Write an integral representing the mass of hay in the pile.
(b) Evaluate the integral.

50. Find the mass M of the solid region W given in spherical coordinates by $0 \le \rho \le 3$, $0 \le \theta < 2\pi$, $0 \le \phi \le \pi/4$. The density, $\delta(P)$, at any point P is given by the distance of P from the origin.

51. Write an integral representing the mass of a sphere of radius 3 if the density of the sphere at any point is twice the distance of that point from the center of the sphere.

52. A sphere is made of material whose density at each point is proportional to the square of the distance of the point from the z-axis. The density is 2 gm/cm^3 at a distance of 2 cm from the axis. What is the mass of the sphere if it is centered at the origin and has radius 3 cm?

53. The density of a solid sphere at any point is proportional to the square of the distance of the point to the center of the sphere. What is the ratio of the mass of a sphere of radius 1 to a sphere of radius 2?

54. A spherical shell centered at the origin has an inner radius of 6 cm and an outer radius of 7 cm. The density, δ, of the material increases linearly with the distance from the center. At the inner surface, $\delta = 9$ gm/cm^3; at the outer surface, $\delta = 11$ gm/cm^3.

(a) Using spherical coordinates, write the density, δ, as a function of radius, ρ.
(b) Write an integral giving the mass of the shell.
(c) Find the mass of the shell.

55. **(a)** Write an iterated integral which represents the mass of a solid ball of radius a. The density at each point in the ball is k times the distance from that point to a fixed plane passing through the center of the ball.
(b) Evaluate the integral.

56. Use appropriate coordinates to find the average distance to the origin for points in the ice cream cone region bounded by the hemisphere $z = \sqrt{8 - x^2 - y^2}$ and the cone $z = \sqrt{x^2 + y^2}$. [Hint: The volume of this region is computed in Problem 27 on page 895.]

For Problems 57–60, use the definition of center of mass given on page 890. Assume x, y, z are in cm.

57. Let C be a solid cone with both height and radius 1 and contained between the surfaces $z = \sqrt{x^2 + y^2}$ and $z = 1$. If C has constant mass density of 1 gm/cm^3, find the z-coordinate of C's center of mass.

58. The density of the cone C in Problem 57 is given by $\delta(z) = z^2$ gm/cm^3. Find

(a) The mass of C.
(b) The z-coordinate of C's center of mass.

59. For $a > 0$, consider the family of solids bounded below by the paraboloid $z = a(x^2 + y^2)$ and above by the plane $z = 1$. If the solids all have constant mass density 1 gm/cm^3, show that the z-coordinate of the center of mass is $2/3$ and so independent of the parameter a.

60. Find the location of the center of mass of a hemisphere of radius a and density b gm/cm^3.

For Problems 61–62, use the definition of moment of inertia given on page 890.

61. The moment of inertia of a solid homogeneous ball B of mass 1 and radius a centered at the origin is the same about any of the coordinate axes (due to the symmetry of the ball). It is easier to evaluate the sum of the three integrals involved in computing the moment of inertia about each of the axes than to compute them individually. Find the sum of the moments of inertia about the x, y and z-axes and thus find the individual moments of inertia.

62. Find the moment of inertia about the z-axis of the solid "fat ice cream cone" given in spherical coordinates by $0 \le \rho \le a$, $0 \le \phi \le \frac{\pi}{3}$ and $0 \le \theta \le 2\pi$. Assume that the solid is homogeneous with mass m.

Problems 63–64 deal with the energy stored in an electric field. If a region of space W contains an electric field whose magnitude at a point (x, y, z) is $E(x, y, z)$, then

$$\text{Energy stored by field} = \frac{1}{2}\int_W \epsilon E^2\, dV,$$

where ϵ is a property of the material called the *permittivity*.

63. The region between two concentric spheres, with radii $a < b$, contains an electric field with magnitude $E = q/(4\pi\epsilon\rho^2)$, where ρ is the distance from the center of the spheres and q is the charge on the inner sphere. Assuming the permittivity, ϵ, is constant, find the total energy stored in the region between the two spheres.

64. Figure 16.53 shows a coaxial cable consisting of two cylindrical conductors centered on the same axis, of radii $a < b$. The electric field between the conductors has magnitude $E = q/(2\pi\epsilon r)$, where r is the distance from the axis and q is the charge per unit length on the cable. The permittivity of the material between the conductors is constant.[4] Show that the stored energy per unit length is proportional to $\ln(b/a)$.

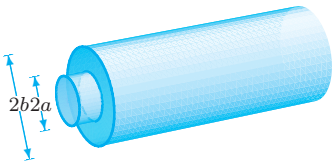


Figure 16.53

65. The density, δ, of a gas in the region under $z = 4 - x^2 - y^2$ and above the xy-plane is $\delta = e^{-x-y}\text{gm/cm}^3$, where x, y, z are in cm. Write an integral, with limits of integration, representing the mass of gas.

66. The density, δ, of the cylinder $x^2 + y^2 \le 4$, $0 \le z \le 3$ varies with the distance, r, from the z-axis:

$$\delta = 1 + r \text{ gm/cm}^3.$$

Find the mass of the cylinder, assuming x, y, z are in cm.

67. The density of material at a point in a solid cylinder is proportional to the distance of the point from the z-axis. What is the ratio of the mass of the cylinder $x^2+y^2 \le 1$, $0 \le z \le 2$ to the mass of the cylinder $x^2 + y^2 \le 9$, $0 \le z \le 2$?

68. A region W consists of the points above the xy-plane and outside the sphere of radius 1 centered at the origin and within the sphere of radius 3 centered at $(0, 0, -1)$. Write an expression for the volume of W. Use cylindrical coordinates and include limits of integration.

69. Compute the force of gravity exerted by a solid cylinder of radius R, height H, and constant density δ on a unit mass at the center of the base of the cylinder.

70. Electric charge is distributed throughout 3-space, with density proportional to the distance from the xy-plane. Show that the total charge inside a cylinder of radius R and height h, sitting on the xy-plane and centered along the z-axis, is proportional to R^2h^2.

71. Electric charge is distributed throughout 3-space with density inversely proportional to the distance from the origin. Show that the total charge inside a sphere of radius R is proportional to R^2.

72. Figure 16.54 shows an alternative notation for spherical coordinates, used often in electrical engineering. Write the volume element dV in this coordinate system.

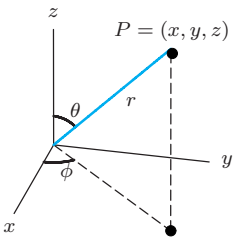


Figure 16.54

Strengthen Your Understanding

73. Which of the following integrals give the volume of the unit sphere?

(a) $\displaystyle\int_0^{2\pi}\int_0^{2\pi}\int_0^{1} d\rho\, d\theta\, d\phi$

(b) $\displaystyle\int_0^{\pi}\int_0^{2\pi}\int_0^{1} d\rho\, d\theta\, d\phi$

(c) $\displaystyle\int_0^{\pi}\int_0^{2\pi}\int_0^{1} \rho^2 \sin\phi\, d\rho\, d\theta\, d\phi$

(d) $\displaystyle\int_0^{\pi}\int_0^{2\pi}\int_0^{1} \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta$

(e) $\displaystyle\int_0^{\pi}\int_0^{2\pi}\int_0^{1} \rho\, d\rho\, d\phi\, d\theta$

In Problems 74–75, explain what is wrong with the statement.

74. The integral $\displaystyle\int_0^{2\pi}\int_0^{\pi}\int_0^{1} 1\, d\rho\, d\phi\, d\theta$ gives the volume inside the sphere of radius 1.

[4] See C. R. Paul and S. A. Nasar, *Introduction to Electromagnetic Fields*, 2nd ed. (New York: McGraw-Hill, 1987).

75. Changing the order of integration gives

$$\int_0^{2\pi}\int_0^{\pi/4}\int_0^{2/\cos\phi} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = \int_0^{2/\cos\phi}\int_0^{\pi/4}\int_0^{2\pi} \rho^2 \sin\phi \, d\theta \, d\phi \, d\rho.$$

In Problems 76–77, give an example of:

76. An integral in spherical coordinates that gives the volume of a hemisphere.

77. An integral for which it is more convenient to use spherical coordinates than to use Cartesian coordinates.

16.6 APPLICATIONS OF INTEGRATION TO PROBABILITY

To represent how a quantity such as height or weight is distributed throughout a population, we use a density function. To study two or more quantities at the same time and see how they are related, we use a multivariable density function.

Density Functions

Distribution of Weight and Height in Expectant Mothers

Table 16.8 shows the distribution of weight and height in a survey of expectant mothers. The histogram in Figure 16.55 is constructed so that the volume of each bar represents the percentage in the corresponding weight and height range. For example, the bar representing the mothers who weighed 60–70 kg and were 160–165 cm tall has base of area $10 \text{ kg} \cdot 5 \text{ cm} = 50 \text{ kg cm}$. The volume of this bar is 12%, so its height is $12\%/50 \text{ kg cm} = 0.24\%/\text{ kg cm}$. Notice that the units on the vertical axis are % per kg cm, so the volume of a bar is a %. The total volume is 100% = 1.

Table 16.8 *Distribution of weight and height in a survey of expectant mothers, in %*

	45-50 kg	50-60 kg	60-70 kg	70-80 kg	80-105 kg	Totals by height
150-155 cm	2	4	4	2	1	13
155-160 cm	0	12	8	2	1	23
160-165 cm	1	7	12	4	3	27
165-170 cm	0	8	12	6	2	28
170-180 cm	0	1	3	4	1	9
Totals by weight	3	32	39	18	8	100

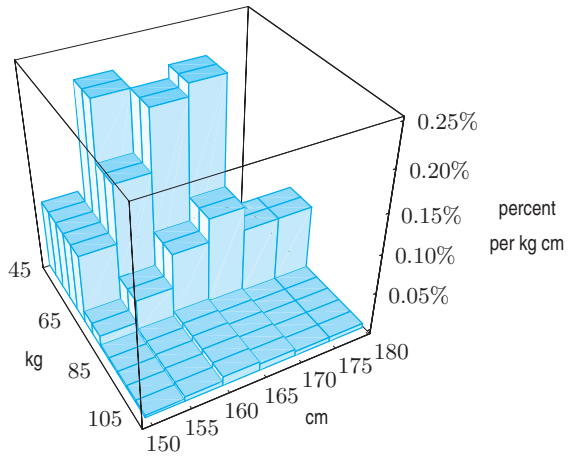


Figure 16.55: Histogram representing the data in Table 16.8